Time limits. The 300-second budget per instance comprises 180 seconds for the exact search and up to 120 seconds for a fallback heuristic. The heuristic may terminate early, so runs without a proof of optimality can finish in slightly more than 180 seconds rather than using the full budget.

1. Detailed Results of Uniform-Weight Instances on S_A under WTFRPA Setting

We first report the results of DK23 on the instance with $ws = 0.3$. Among these instances, only the instance "60-10-sif-0.3-3" is feasible.

Table 1 Detailed Results of DK23 on 60-10-sif-0.3-3

ws-0.3			
Name	Cost	Gap‰	CPU
60-10-sif-3	1081.0	0.00	0.01

1.1. Results on the instances with i execution mode

Table 2 Detailed Results of DK23 on Uniform-Weight Instances in S_A with i execution mode ($ws = 0.4 - 0.6$)

ws-0.4				ws-0.5				ws-0.6			
Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU
60-10-i-1	-	-	-	60-10-i-1	-	-	-	60-10-i-1	-	-	-
60-10-i-2	-	-	-	60-10-i-2	-	-	-	60-10-i-2	-	-	-
60-10-i-3	-	-	-	60-10-i-3	-	-	-	60-10-i-3	1082.0	0.00	0.18
60-10-i-4	-	-	-	60-10-i-4	-	-	-	60-10-i-4	-	-	-
60-10-i-5	-	-	-	60-10-i-5	-	-	-	60-10-i-5	-	-	-
60-20-i-1	-	-	-	60-20-i-1	-	-	-	60-20-i-1	-	-	-
60-20-i-2	-	-	-	60-20-i-2	-	-	-	60-20-i-2	-	-	-
60-20-i-3	-	-	-	60-20-i-3	-	-	-	60-20-i-3	-	-	-
60-20-i-4	-	-	-	60-20-i-4	-	-	-	60-20-i-4	-	-	-
60-20-i-5	-	-	-	60-20-i-5	-	-	-	60-20-i-5	-	-	-
60-30-i-1	-	-	-	60-30-i-1	-	-	-	60-30-i-1	-	-	-
60-30-i-2	-	-	-	60-30-i-2	-	-	-	60-30-i-2	-	-	-
60-30-i-3	-	-	-	60-30-i-3	-	-	-	60-30-i-3	-	-	-
60-30-i-4	-	-	-	60-30-i-4	-	-	-	60-30-i-4	-	-	-
60-30-i-5	-	-	-	60-30-i-5	-	-	-	60-30-i-5	-	-	-
90-10-i-1	-	-	-	90-10-i-1	-	-	-	90-10-i-1	-	-	-
90-10-i-2	-	-	-	90-10-i-2	-	-	-	90-10-i-2	-	-	-
90-10-i-3	-	-	-	90-10-i-3	-	-	-	90-10-i-3	-	-	-
90-10-i-4	-	-	-	90-10-i-4	-	-	-	90-10-i-4	-	-	-
90-10-i-5	-	-	-	90-10-i-5	-	-	-	90-10-i-5	-	-	-
90-20-i-1	-	-	-	90-20-i-1	-	-	-	90-20-i-1	-	-	-
90-20-i-2	-	-	-	90-20-i-2	-	-	-	90-20-i-2	-	-	-
90-20-i-3	-	-	-	90-20-i-3	-	-	-	90-20-i-3	-	-	-
90-20-i-4	-	-	-	90-20-i-4	-	-	-	90-20-i-4	-	-	-
90-20-i-5	-	-	-	90-20-i-5	-	-	-	90-20-i-5	-	-	-
90-30-i-1	-	-	-	90-30-i-1	-	-	-	90-30-i-1	-	-	-
90-30-i-2	-	-	-	90-30-i-2	-	-	-	90-30-i-2	-	-	-
90-30-i-3	-	-	-	90-30-i-3	-	-	-	90-30-i-3	-	-	-
90-30-i-4	-	-	-	90-30-i-4	-	-	-	90-30-i-4	-	-	-
90-30-i-5	-	-	-	90-30-i-5	-	-	-	90-30-i-5	-	-	-
120-10-i-1	-	-	-	120-10-i-1	-	-	-	120-10-i-1	-	-	-
120-10-i-2	-	-	-	120-10-i-2	-	-	-	120-10-i-2	-	-	-
120-10-i-3	-	-	-	120-10-i-3	-	-	-	120-10-i-3	-	-	-
120-10-i-4	-	-	-	120-10-i-4	-	-	-	120-10-i-4	-	-	-
120-10-i-5	-	-	-	120-10-i-5	-	-	-	120-10-i-5	-	-	-
120-20-i-1	-	-	-	120-20-i-1	-	-	-	120-20-i-1	-	-	-
120-20-i-2	-	-	-	120-20-i-2	-	-	-	120-20-i-2	-	-	-
120-20-i-3	-	-	-	120-20-i-3	-	-	-	120-20-i-3	-	-	-
120-20-i-4	-	-	-	120-20-i-4	-	-	-	120-20-i-4	-	-	-
120-20-i-5	-	-	-	120-20-i-5	-	-	-	120-20-i-5	-	-	-
120-30-i-1	-	-	-	120-30-i-1	-	-	-	120-30-i-1	-	-	-
120-30-i-2	-	-	-	120-30-i-2	-	-	-	120-30-i-2	6796.0	7.50	180.15
120-30-i-3	-	-	-	120-30-i-3	-	-	-	120-30-i-3	6689.0	0.00	36.46
120-30-i-4	-	-	-	120-30-i-4	-	-	-	120-30-i-4	-	-	-
120-30-i-5	-	-	-	120-30-i-5	-	-	-	120-30-i-5	-	-	-

Table 3 Detailed Results of DK23 on Uniform-Weight Instances in S_A with i execution mode ($ws = 0.7 - 0.9$)

ws-0.7				ws-0.8				ws-0.9			
Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU
60-10-i-1	-	-	-	60-10-i-1	1043.0	0.00	0.01	60-10-i-1	1042.0	0.00	0.02
60-10-i-2	1062.0	0.00	1.10	60-10-i-2	1059.0	0.00	0.02	60-10-i-2	1056.0	0.00	0.01
60-10-i-3	1077.0	0.00	0.02	60-10-i-3	1077.0	0.00	0.01	60-10-i-3	1077.0	0.00	0.03
60-10-i-4	-	-	-	60-10-i-4	-	-	-	60-10-i-4	-	-	-
60-10-i-5	-	-	-	60-10-i-5	-	-	-	60-10-i-5	1066.0	0.00	0.01
60-20-i-1	2152.0	0.00	0.04	60-20-i-1	2150.0	0.00	0.01	60-20-i-1	2150.0	0.00	0.01
60-20-i-2	2134.0	0.00	0.19	60-20-i-2	2130.0	0.00	0.06	60-20-i-2	2129.0	0.00	0.01
60-20-i-3	2028.0	0.00	0.17	60-20-i-3	2026.0	0.00	0.01	60-20-i-3	2025.0	0.00	0.02
60-20-i-4	-	-	-	60-20-i-4	-	-	-	60-20-i-4	2026.0	0.00	0.02
60-20-i-5	-	-	-	60-20-i-5	2150.0	0.00	0.01	60-20-i-5	2150.0	0.00	0.01
60-30-i-1	-	-	-	60-30-i-1	3130.0	0.00	0.05	60-30-i-1	3128.0	0.00	0.04
60-30-i-2	-	-	-	60-30-i-2	-	-	-	60-30-i-2	3204.0	0.00	0.02
60-30-i-3	-	-	-	60-30-i-3	3185.0	0.00	0.03	60-30-i-3	3184.0	0.00	0.10
60-30-i-4	-	-	-	60-30-i-4	3183.0	0.00	0.05	60-30-i-4	3181.0	0.00	0.05
60-30-i-5	-	-	-	60-30-i-5	-	-	-	60-30-i-5	3191.0	0.00	0.04
90-10-i-1	-	-	-	90-10-i-1	1636.0	0.00	0.04	90-10-i-1	1636.0	0.00	0.02
90-10-i-2	-	-	-	90-10-i-2	1677.0	0.00	0.05	90-10-i-2	1677.0	0.00	0.14
90-10-i-3	1626.0	0.00	0.02	90-10-i-3	1626.0	0.00	0.07	90-10-i-3	1624.0	0.00	0.02
90-10-i-4	-	-	-	90-10-i-4	-	-	-	90-10-i-4	1649.0	0.00	0.02
90-10-i-5	-	-	-	90-10-i-5	-	-	-	90-10-i-5	1644.0	0.00	0.02
90-20-i-1	-	-	-	90-20-i-1	3307.0	0.00	0.13	90-20-i-1	3307.0	0.00	0.03
90-20-i-2	-	-	-	90-20-i-2	3329.0	0.00	0.20	90-20-i-2	3327.0	0.00	0.04
90-20-i-3	-	-	-	90-20-i-3	3330.0	0.00	0.76	90-20-i-3	3327.0	0.00	0.11
90-20-i-4	3322.0	4.89	180.02	90-20-i-4	3317.0	0.00	0.06	90-20-i-4	3315.0	0.00	0.10
90-20-i-5	-	-	-	90-20-i-5	3288.0	0.00	0.16	90-20-i-5	3286.0	0.00	0.13
90-30-i-1	-	-	-	90-30-i-1	4948.0	0.00	0.22	90-30-i-1	4948.0	0.00	0.08
90-30-i-2	-	-	-	90-30-i-2	-	-	-	90-30-i-2	4835.0	0.00	0.07
90-30-i-3	-	-	-	90-30-i-3	-	-	-	90-30-i-3	5012.0	0.00	0.04
90-30-i-4	-	-	-	90-30-i-4	4720.0	0.00	0.17	90-30-i-4	4720.0	0.00	0.07
90-30-i-5	-	-	-	90-30-i-5	4928.0	0.00	0.28	90-30-i-5	4928.0	0.00	0.05
120-10-i-1	2214.0	13.30	180.02	120-10-i-1	2213.0	0.00	30.70	120-10-i-1	2209.0	0.00	0.16
120-10-i-2	-	-	-	120-10-i-2	-	-	-	120-10-i-2	2216.0	0.00	0.03
120-10-i-3	-	-	-	120-10-i-3	-	-	-	120-10-i-3	2223.0	0.00	0.05
120-10-i-4	-	-	-	120-10-i-4	2176.0	0.00	0.07	120-10-i-4	2176.0	0.00	0.07
120-10-i-5	-	-	-	120-10-i-5	-	-	-	120-10-i-5	2253.0	0.00	0.07
120-20-i-1	-	-	-	120-20-i-1	-	-	-	120-20-i-1	4583.0	0.00	0.27
120-20-i-2	-	-	-	120-20-i-2	-	-	-	120-20-i-2	4541.0	0.00	0.08
120-20-i-3	-	-	-	120-20-i-3	4520.0	0.00	0.08	120-20-i-3	4520.0	0.00	0.51
120-20-i-4	-	-	-	120-20-i-4	-	-	-	120-20-i-4	4431.0	0.00	0.07
120-20-i-5	4486.0	0.00	180.03	120-20-i-5	4485.0	0.00	1.00	120-20-i-5	4485.0	0.00	0.04
120-30-i-1	6724.0	2.97	180.09	120-30-i-1	6720.0	1.94	180.08	120-30-i-1	6717.0	0.00	0.55
120-30-i-2	6786.0	0.00	0.32	120-30-i-2	6784.0	0.00	1.08	120-30-i-2	6784.0	0.00	0.26
120-30-i-3	6685.0	0.00	0.32	120-30-i-3	6685.0	0.00	0.73	120-30-i-3	6685.0	0.00	0.70
120-30-i-4	-	-	-	120-30-i-4	6703.0	0.00	0.12	120-30-i-4	6700.0	0.00	0.15
120-30-i-5	-	-	-	120-30-i-5	-	-	-	120-30-i-5	6635.0	0.00	2.31

1.2. Results on the instances with sf execution mode

Table 4 Detailed Results of DK23 on Uniform-Weight Instances in S_A with sf execution mode ($ws = 0.4 - 0.6$)

ws-0.4				ws-0.5				ws-0.6			
Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU
60-10-sf-1	1044.0	0.00	0.12	60-10-sf-1	1036.0	0.00	0.10	60-10-sf-1	1034.0	0.00	0.54
60-10-sf-2	-	-	-	60-10-sf-2	-	-	-	60-10-sf-2	1052.0	0.00	1.60
60-10-sf-3	1075.0	0.00	0.23	60-10-sf-3	1070.0	0.00	0.06	60-10-sf-3	1069.0	0.00	0.13
60-10-sf-4	-	-	-	60-10-sf-4	1063.0	0.00	0.02	60-10-sf-4	1057.0	0.00	0.06
60-10-sf-5	-	-	-	60-10-sf-5	1063.0	0.00	0.04	60-10-sf-5	1058.0	0.00	0.02
60-20-sf-1	-	-	-	60-20-sf-1	2142.0	0.00	0.04	60-20-sf-1	2137.0	0.00	0.11
60-20-sf-2	-	-	-	60-20-sf-2	-	-	-	60-20-sf-2	2126.0	0.00	0.16
60-20-sf-3	-	-	-	60-20-sf-3	-	-	-	60-20-sf-3	-	-	-
60-20-sf-4	-	-	-	60-20-sf-4	2018.0	0.00	0.10	60-20-sf-4	2011.0	0.00	0.04
60-20-sf-5	-	-	-	60-20-sf-5	2147.0	0.00	0.05	60-20-sf-5	2139.0	0.00	0.15
60-30-sf-1	-	-	-	60-30-sf-1	3108.0	0.00	0.28	60-30-sf-1	3099.0	0.00	0.08
60-30-sf-2	-	-	-	60-30-sf-2	-	-	-	60-30-sf-2	3183.0	0.00	180.71
60-30-sf-3	-	-	-	60-30-sf-3	-	-	-	60-30-sf-3	3171.0	0.00	0.17
60-30-sf-4	-	-	-	60-30-sf-4	3177.0	0.00	0.18	60-30-sf-4	3162.0	0.00	1.07
60-30-sf-5	-	-	-	60-30-sf-5	-	-	-	60-30-sf-5	3172.0	0.00	0.41
90-10-sf-1	-	-	-	90-10-sf-1	-	-	-	90-10-sf-1	1623.0	0.00	0.89
90-10-sf-2	-	-	-	90-10-sf-2	-	-	-	90-10-sf-2	1666.0	0.00	0.02
90-10-sf-3	-	-	-	90-10-sf-3	1614.0	0.00	0.03	90-10-sf-3	1611.0	0.00	0.02
90-10-sf-4	-	-	-	90-10-sf-4	-	-	-	90-10-sf-4	-	-	-
90-10-sf-5	1642.0	6.09	180.06	90-10-sf-5	1636.0	0.00	0.12	90-10-sf-5	1632.0	0.00	0.17
90-20-sf-1	-	-	-	90-20-sf-1	3291.0	0.00	0.25	90-20-sf-1	3281.0	0.00	3.29
90-20-sf-2	-	-	-	90-20-sf-2	3305.0	0.00	53.94	90-20-sf-2	3295.0	3.54	180.10
90-20-sf-3	-	-	-	90-20-sf-3	3308.0	0.00	1.26	90-20-sf-3	3298.0	0.00	1.08
90-20-sf-4	-	-	-	90-20-sf-4	-	-	-	90-20-sf-4	3301.0	0.00	3.30
90-20-sf-5	-	-	-	90-20-sf-5	-	-	-	90-20-sf-5	3262.0	3.07	180.06
90-30-sf-1	-	-	-	90-30-sf-1	-	-	-	90-30-sf-1	-	-	-
90-30-sf-2	-	-	-	90-30-sf-2	-	-	-	90-30-sf-2	-	-	-
90-30-sf-3	-	-	-	90-30-sf-3	4989.0	3.67	180.81	90-30-sf-3	4977.0	0.00	10.77
90-30-sf-4	-	-	-	90-30-sf-4	4705.0	0.00	12.12	90-30-sf-4	4683.0	0.00	9.91
90-30-sf-5	-	-	-	90-30-sf-5	4903.0	0.00	102.92	90-30-sf-5	4891.0	2.04	180.25
120-10-sf-1	-	-	-	120-10-sf-1	-	-	-	120-10-sf-1	2197.0	4.55	180.09
120-10-sf-2	-	-	-	120-10-sf-2	-	-	-	120-10-sf-2	2206.0	0.00	0.07
120-10-sf-3	-	-	-	120-10-sf-3	-	-	-	120-10-sf-3	2208.0	0.00	0.09
120-10-sf-4	-	-	-	120-10-sf-4	2158.0	0.00	0.79	120-10-sf-4	2157.0	0.00	0.22
120-10-sf-5	-	-	-	120-10-sf-5	-	-	-	120-10-sf-5	2239.0	0.00	0.14
120-20-sf-1	4558.0	13.14	180.08	120-20-sf-1	4548.0	13.11	180.24	120-20-sf-1	4538.0	0.00	162.83
120-20-sf-2	-	-	-	120-20-sf-2	4520.0	7.07	180.25	120-20-sf-2	4506.0	0.00	180.15
120-20-sf-3	-	-	-	120-20-sf-3	-	-	-	120-20-sf-3	4479.0	2.56	180.20
120-20-sf-4	-	-	-	120-20-sf-4	-	-	-	120-20-sf-4	-	-	-
120-20-sf-5	-	-	-	120-20-sf-5	-	-	-	120-20-sf-5	-	-	-
120-30-sf-1	-	-	-	120-30-sf-1	6686.0	5.48	180.21	120-30-sf-1	6667.0	9.77	180.47
120-30-sf-2	-	-	-	120-30-sf-2	6752.0	3.76	181.38	120-30-sf-2	6734.0	2.35	180.76
120-30-sf-3	-	-	-	120-30-sf-3	6670.0	35.52	180.79	120-30-sf-3	6636.0	0.00	15.09
120-30-sf-4	-	-	-	120-30-sf-4	6676.0	6.11	182.08	120-30-sf-4	6654.0	2.86	180.95
120-30-sf-5	-	-	-	120-30-sf-5	-	-	-	120-30-sf-5	-	-	-

Table 5 Detailed Results of DK23 on Uniform-Weight Instances in S_A with sf execution mode ($ws = 0.7 - 0.9$)

ws-0.7				ws-0.8				ws-0.9			
Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU
60-10-sf-1	1031.0	0.00	0.07	60-10-sf-1	1031.0	0.00	0.03	60-10-sf-1	1031.0	0.00	0.03
60-10-sf-2	1049.0	0.00	0.04	60-10-sf-2	1049.0	0.00	0.05	60-10-sf-2	1048.0	0.00	0.01
60-10-sf-3	1067.0	0.00	0.04	60-10-sf-3	1063.0	0.00	0.02	60-10-sf-3	1063.0	0.00	0.02
60-10-sf-4	1053.0	0.00	0.05	60-10-sf-4	1052.0	0.00	0.04	60-10-sf-4	1051.0	0.00	0.02
60-10-sf-5	1056.0	0.00	0.02	60-10-sf-5	1055.0	0.00	0.04	60-10-sf-5	1053.0	0.00	0.03
60-20-sf-1	2134.0	0.00	0.26	60-20-sf-1	2132.0	0.00	0.02	60-20-sf-1	2132.0	0.00	0.02
60-20-sf-2	2116.0	0.00	0.03	60-20-sf-2	2110.0	0.00	0.05	60-20-sf-2	2109.0	0.00	0.03
60-20-sf-3	2018.0	0.00	0.18	60-20-sf-3	2017.0	0.00	0.05	60-20-sf-3	2014.0	0.00	0.02
60-20-sf-4	2009.0	0.00	0.16	60-20-sf-4	2007.0	0.00	0.17	60-20-sf-4	2006.0	0.00	0.09
60-20-sf-5	2134.0	0.00	0.21	60-20-sf-5	2133.0	0.00	0.11	60-20-sf-5	2132.0	0.00	0.09
60-30-sf-1	3096.0	0.00	0.13	60-30-sf-1	3093.0	0.00	0.16	60-30-sf-1	3092.0	0.00	0.09
60-30-sf-2	3169.0	0.00	1.94	60-30-sf-2	3166.0	0.00	0.87	60-30-sf-2	3165.0	0.00	0.07
60-30-sf-3	3162.0	0.00	0.07	60-30-sf-3	3157.0	0.00	0.31	60-30-sf-3	3155.0	0.00	0.55
60-30-sf-4	3156.0	0.00	1.77	60-30-sf-4	3152.0	0.00	0.48	60-30-sf-4	3150.0	0.00	0.37
60-30-sf-5	3164.0	0.00	0.39	60-30-sf-5	3162.0	0.00	6.05	60-30-sf-5	3160.0	0.00	0.21
90-10-sf-1	1622.0	0.00	152.52	90-10-sf-1	1619.0	9.26	180.02	90-10-sf-1	1619.0	9.26	180.02
90-10-sf-2	1662.0	0.00	0.44	90-10-sf-2	1659.0	0.00	0.28	90-10-sf-2	1658.0	0.00	0.01
90-10-sf-3	1610.0	0.00	0.02	90-10-sf-3	1610.0	0.00	0.01	90-10-sf-3	1610.0	0.00	0.02
90-10-sf-4	-	-	-	90-10-sf-4	1635.0	0.00	0.05	90-10-sf-4	1635.0	0.00	0.07
90-10-sf-5	1630.0	0.00	0.12	90-10-sf-5	1629.0	0.00	0.03	90-10-sf-5	1629.0	0.00	0.08
90-20-sf-1	3275.0	0.00	0.26	90-20-sf-1	3273.0	0.00	0.15	90-20-sf-1	3273.0	0.00	0.03
90-20-sf-2	3293.0	4.56	180.12	90-20-sf-2	3291.0	3.04	180.09	90-20-sf-2	3289.0	0.00	0.08
90-20-sf-3	3296.0	0.00	11.54	90-20-sf-3	3295.0	0.00	0.44	90-20-sf-3	3295.0	0.00	0.50
90-20-sf-4	3292.0	0.00	0.71	90-20-sf-4	3288.0	0.00	0.52	90-20-sf-4	3286.0	0.00	0.24
90-20-sf-5	3257.0	3.07	180.07	90-20-sf-5	3251.0	0.00	0.43	90-20-sf-5	3249.0	0.00	0.05
90-30-sf-1	4903.0	0.00	10.08	90-30-sf-1	4891.0	0.00	4.03	90-30-sf-1	4889.0	0.00	0.35
90-30-sf-2	4801.0	4.69	180.11	90-30-sf-2	4795.0	4.17	180.09	90-30-sf-2	4791.0	0.00	50.78
90-30-sf-3	4970.0	0.00	8.80	90-30-sf-3	4969.0	0.00	1.03	90-30-sf-3	4966.0	0.00	0.89
90-30-sf-4	4675.0	2.14	180.19	90-30-sf-4	4673.0	0.00	0.69	90-30-sf-4	4672.0	0.00	0.27
90-30-sf-5	4886.0	0.00	1.35	90-30-sf-5	4885.0	0.00	2.29	90-30-sf-5	4884.0	0.00	0.11
120-10-sf-1	2194.0	0.00	1.72	120-10-sf-1	2192.0	0.00	0.04	120-10-sf-1	2192.0	0.00	0.02
120-10-sf-2	2206.0	0.00	0.05	120-10-sf-2	2204.0	0.00	0.04	120-10-sf-2	2204.0	0.00	0.04
120-10-sf-3	2206.0	0.00	1.71	120-10-sf-3	2205.0	5.48	180.06	120-10-sf-3	2205.0	5.67	180.07
120-10-sf-4	2156.0	0.00	0.07	120-10-sf-4	2155.0	0.00	0.29	120-10-sf-4	2150.0	0.00	0.07
120-10-sf-5	2237.0	0.00	0.13	120-10-sf-5	2235.0	0.00	0.70	120-10-sf-5	2235.0	0.00	4.17
120-20-sf-1	4536.0	0.00	0.30	120-20-sf-1	4533.0	0.00	0.07	120-20-sf-1	4533.0	0.00	0.61
120-20-sf-2	4498.0	0.00	0.16	120-20-sf-2	4495.0	0.00	0.96	120-20-sf-2	4492.0	0.00	0.32
120-20-sf-3	4470.0	5.07	180.48	120-20-sf-3	4464.0	0.00	180.11	120-20-sf-3	4461.0	0.00	29.12
120-20-sf-4	-	-	-	120-20-sf-4	4386.0	0.00	0.41	120-20-sf-4	4385.0	0.00	0.49
120-20-sf-5	-	-	-	120-20-sf-5	4446.0	0.00	0.95	120-20-sf-5	4446.0	0.00	0.29
120-30-sf-1	6656.0	0.00	0.28	120-30-sf-1	6655.0	0.00	0.38	120-30-sf-1	6655.0	0.00	6.96
120-30-sf-2	6727.0	2.60	180.53	120-30-sf-2	6724.0	0.00	180.30	120-30-sf-2	6722.0	0.00	0.93
120-30-sf-3	6633.0	0.00	0.59	120-30-sf-3	6633.0	0.00	1.80	120-30-sf-3	6633.0	0.00	1.59
120-30-sf-4	6640.0	0.00	180.20	120-30-sf-4	6639.0	1.51	180.45	120-30-sf-4	6636.0	0.00	1.55
120-30-sf-5	6568.0	0.00	40.14	120-30-sf-5	6565.0	0.00	0.58	120-30-sf-5	6558.0	0.00	1.46

1.3. Results on the instances with sif execution mode

Table 6 Detailed Results of DK23 on Uniform-Weight Instances in S_A with sif execution mode ($ws = 0.4 - 0.6$)

ws-0.4				ws-0.5				ws-0.6			
Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU
60-10-sif-1	1042.0	0.00	0.01	60-10-sif-1	1038.0	0.00	0.04	60-10-sif-1	1035.0	0.00	0.07
60-10-sif-2	-	-	-	60-10-sif-2	1063.0	0.00	0.03	60-10-sif-2	1054.0	0.00	0.02
60-10-sif-3	1076.0	0.00	0.07	60-10-sif-3	1071.0	0.00	0.08	60-10-sif-3	1069.0	0.00	0.18
60-10-sif-4	-	-	-	60-10-sif-4	1063.0	0.00	0.02	60-10-sif-4	1057.0	0.00	0.03
60-10-sif-5	1062.0	0.00	0.01	60-10-sif-5	1059.0	0.00	0.12	60-10-sif-5	1057.0	0.00	0.03
60-20-sif-1	-	-	-	60-20-sif-1	2142.0	0.00	0.03	60-20-sif-1	2137.0	0.00	0.15
60-20-sif-2	-	-	-	60-20-sif-2	2126.0	0.00	0.20	60-20-sif-2	2120.0	4.72	180.06
60-20-sif-3	-	-	-	60-20-sif-3	-	-	-	60-20-sif-3	2024.0	0.00	1.58
60-20-sif-4	-	-	-	60-20-sif-4	2017.0	0.00	0.15	60-20-sif-4	2011.0	0.00	0.05
60-20-sif-5	-	-	-	60-20-sif-5	2146.0	0.00	0.24	60-20-sif-5	2139.0	0.00	0.71
60-30-sif-1	-	-	-	60-30-sif-1	3105.0	0.00	2.74	60-30-sif-1	3101.0	0.00	1.39
60-30-sif-2	-	-	-	60-30-sif-2	-	-	-	60-30-sif-2	3184.0	0.00	15.15
60-30-sif-3	-	-	-	60-30-sif-3	-	-	-	60-30-sif-3	3166.0	0.00	0.29
60-30-sif-4	-	-	-	60-30-sif-4	-	-	-	60-30-sif-4	3161.0	0.00	0.33
60-30-sif-5	-	-	-	60-30-sif-5	-	-	-	60-30-sif-5	3175.0	0.00	22.14
90-10-sif-1	-	-	-	90-10-sif-1	1629.0	0.00	30.02	90-10-sif-1	1624.0	0.00	4.31
90-10-sif-2	-	-	-	90-10-sif-2	1673.0	0.00	0.37	90-10-sif-2	1663.0	0.00	0.03
90-10-sif-3	-	-	-	90-10-sif-3	1617.0	0.00	0.02	90-10-sif-3	1612.0	0.00	0.02
90-10-sif-4	-	-	-	90-10-sif-4	-	-	-	90-10-sif-4	-	-	-
90-10-sif-5	-	-	-	90-10-sif-5	1636.0	0.00	0.29	90-10-sif-5	1632.0	0.00	0.33
90-20-sif-1	-	-	-	90-20-sif-1	3295.0	0.00	0.32	90-20-sif-1	3281.0	0.00	9.90
90-20-sif-2	-	-	-	90-20-sif-2	3317.0	10.37	180.51	90-20-sif-2	3300.0	0.00	5.91
90-20-sif-3	-	-	-	90-20-sif-3	3310.0	0.00	13.45	90-20-sif-3	3299.0	0.00	0.94
90-20-sif-4	-	-	-	90-20-sif-4	-	-	-	90-20-sif-4	3300.0	0.00	0.40
90-20-sif-5	-	-	-	90-20-sif-5	-	-	-	90-20-sif-5	3262.0	0.00	90.50
90-30-sif-1	-	-	-	90-30-sif-1	4934.0	8.93	182.25	90-30-sif-1	4912.0	4.17	181.44
90-30-sif-2	-	-	-	90-30-sif-2	-	-	-	90-30-sif-2	4810.0	0.00	180.15
90-30-sif-3	-	-	-	90-30-sif-3	4992.0	5.04	181.49	90-30-sif-3	4978.0	0.00	7.25
90-30-sif-4	-	-	-	90-30-sif-4	4698.0	0.00	19.58	90-30-sif-4	4681.0	0.00	25.94
90-30-sif-5	4932.0	0.00	0.49	90-30-sif-5	4902.0	0.00	42.25	90-30-sif-5	4890.0	0.00	7.40
120-10-sif-1	-	-	-	120-10-sif-1	-	-	-	120-10-sif-1	2200.0	6.82	180.20
120-10-sif-2	2214.0	0.00	12.67	120-10-sif-2	2207.0	0.00	0.03	120-10-sif-2	2206.0	0.00	0.03
120-10-sif-3	-	-	-	120-10-sif-3	-	-	-	120-10-sif-3	2209.0	0.00	28.18
120-10-sif-4	-	-	-	120-10-sif-4	2159.0	0.00	12.03	120-10-sif-4	2155.0	0.00	0.31
120-10-sif-5	-	-	-	120-10-sif-5	-	-	-	120-10-sif-5	2240.0	0.00	0.55
120-20-sif-1	-	-	-	120-20-sif-1	4546.0	2.89	180.38	120-20-sif-1	4538.0	0.00	180.18
120-20-sif-2	-	-	-	120-20-sif-2	4519.0	0.00	0.85	120-20-sif-2	4503.0	0.00	0.33
120-20-sif-3	-	-	-	120-20-sif-3	-	-	-	120-20-sif-3	4482.0	0.00	19.58
120-20-sif-4	-	-	-	120-20-sif-4	4410.0	9.55	181.03	120-20-sif-4	4397.0	0.00	14.60
120-20-sif-5	-	-	-	120-20-sif-5	-	-	-	120-20-sif-5	4457.0	0.00	4.67
120-30-sif-1	-	-	-	120-30-sif-1	6706.0	7.10	180.85	120-30-sif-1	6666.0	1.50	180.67
120-30-sif-2	-	-	-	120-30-sif-2	6749.0	3.79	183.37	120-30-sif-2	6733.0	1.52	180.72
120-30-sif-3	-	-	-	120-30-sif-3	6656.0	2.76	181.48	120-30-sif-3	6636.0	0.00	23.92
120-30-sif-4	-	-	-	120-30-sif-4	6684.0	4.61	181.02	120-30-sif-4	6652.0	2.76	181.83
120-30-sif-5	-	-	-	120-30-sif-5	-	-	-	120-30-sif-5	6579.0	1.68	180.62

Table 7 Detailed Results of DK23 on Uniform-Weight Instances in S_A with sif execution mode ($ws = 0.7 - 0.9$)

ws-0.7				ws-0.8				ws-0.9			
Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU
60-10-sif-1	1033.0	0.00	0.08	60-10-sif-1	1031.0	0.00	0.04	60-10-sif-1	1031.0	0.00	0.03
60-10-sif-2	1053.0	0.00	0.06	60-10-sif-2	1049.0	0.00	0.05	60-10-sif-2	1048.0	0.00	0.02
60-10-sif-3	1068.0	0.00	0.08	60-10-sif-3	1066.0	0.00	0.02	60-10-sif-3	1063.0	0.00	0.01
60-10-sif-4	1053.0	0.00	0.04	60-10-sif-4	1052.0	0.00	0.04	60-10-sif-4	1051.0	0.00	0.02
60-10-sif-5	1055.0	0.00	0.09	60-10-sif-5	1054.0	0.00	0.03	60-10-sif-5	1053.0	0.00	0.03
60-20-sif-1	2134.0	0.00	0.04	60-20-sif-1	2132.0	0.00	0.03	60-20-sif-1	2132.0	0.00	0.03
60-20-sif-2	2115.0	0.00	0.09	60-20-sif-2	2113.0	0.00	0.20	60-20-sif-2	2109.0	0.00	0.05
60-20-sif-3	2016.0	0.00	0.02	60-20-sif-3	2016.0	0.00	0.06	60-20-sif-3	2014.0	0.00	0.03
60-20-sif-4	2010.0	0.00	0.30	60-20-sif-4	2007.0	0.00	0.23	60-20-sif-4	2006.0	0.00	0.15
60-20-sif-5	2134.0	0.00	0.10	60-20-sif-5	2133.0	0.00	0.15	60-20-sif-5	2132.0	0.00	0.02
60-30-sif-1	3096.0	0.00	2.39	60-30-sif-1	3091.0	0.00	0.12	60-30-sif-1	3090.0	0.00	0.09
60-30-sif-2	3171.0	0.00	1.45	60-30-sif-2	3166.0	0.00	1.48	60-30-sif-2	3165.0	0.00	0.07
60-30-sif-3	3161.0	0.00	0.15	60-30-sif-3	3158.0	0.00	0.11	60-30-sif-3	3155.0	0.00	0.05
60-30-sif-4	3155.0	0.00	4.00	60-30-sif-4	3152.0	0.00	0.62	60-30-sif-4	3150.0	0.00	0.30
60-30-sif-5	3165.0	0.00	32.08	60-30-sif-5	3162.0	0.00	1.66	60-30-sif-5	3160.0	0.00	0.13
90-10-sif-1	1621.0	0.00	7.37	90-10-sif-1	1620.0	9.26	180.04	90-10-sif-1	1617.0	0.00	0.03
90-10-sif-2	1660.0	0.00	0.14	90-10-sif-2	1659.0	0.00	0.12	90-10-sif-2	1658.0	0.00	0.02
90-10-sif-3	1610.0	0.00	0.02	90-10-sif-3	1610.0	0.00	0.08	90-10-sif-3	1610.0	0.00	0.02
90-10-sif-4	1637.0	0.00	0.03	90-10-sif-4	1637.0	0.00	0.07	90-10-sif-4	1635.0	0.00	0.26
90-10-sif-5	1631.0	0.00	0.15	90-10-sif-5	1629.0	0.00	0.03	90-10-sif-5	1629.0	0.00	0.11
90-20-sif-1	3275.0	0.00	1.53	90-20-sif-1	3273.0	0.00	0.34	90-20-sif-1	3273.0	0.00	0.10
90-20-sif-2	3293.0	4.56	180.13	90-20-sif-2	3291.0	3.04	180.09	90-20-sif-2	3289.0	0.00	0.32
90-20-sif-3	3296.0	0.00	0.63	90-20-sif-3	3293.0	0.00	0.22	90-20-sif-3	3293.0	0.00	0.13
90-20-sif-4	3292.0	0.00	1.26	90-20-sif-4	3288.0	0.00	0.87	90-20-sif-4	3286.0	0.00	1.41
90-20-sif-5	3255.0	0.00	2.02	90-20-sif-5	3251.0	0.00	0.59	90-20-sif-5	3249.0	0.00	0.32
90-30-sif-1	4900.0	0.00	61.66	90-30-sif-1	4898.0	0.00	0.62	90-30-sif-1	4889.0	0.00	0.26
90-30-sif-2	4798.0	0.00	180.13	90-30-sif-2	4795.0	4.17	180.17	90-30-sif-2	4793.0	2.61	180.12
90-30-sif-3	4972.0	0.00	3.74	90-30-sif-3	4969.0	0.00	2.56	90-30-sif-3	4966.0	0.00	0.34
90-30-sif-4	4675.0	0.00	2.71	90-30-sif-4	4673.0	0.00	0.71	90-30-sif-4	4672.0	0.00	0.50
90-30-sif-5	4886.0	0.00	3.28	90-30-sif-5	4885.0	0.00	11.88	90-30-sif-5	4884.0	0.00	1.03
120-10-sif-1	2198.0	9.10	180.13	120-10-sif-1	2192.0	0.00	0.05	120-10-sif-1	2192.0	0.00	0.16
120-10-sif-2	2205.0	0.00	3.83	120-10-sif-2	2203.0	0.00	0.03	120-10-sif-2	2203.0	0.00	0.06
120-10-sif-3	2205.0	0.00	0.03	120-10-sif-3	2204.0	0.00	0.07	120-10-sif-3	2204.0	0.00	0.58
120-10-sif-4	2155.0	0.00	0.14	120-10-sif-4	2154.0	0.00	0.11	120-10-sif-4	2150.0	0.00	0.09
120-10-sif-5	2236.0	0.00	0.17	120-10-sif-5	2234.0	0.00	0.09	120-10-sif-5	2234.0	0.00	0.12
120-20-sif-1	4536.0	0.00	0.42	120-20-sif-1	4535.0	0.00	0.34	120-20-sif-1	4533.0	0.00	0.11
120-20-sif-2	4498.0	0.00	0.48	120-20-sif-2	4495.0	0.00	0.12	120-20-sif-2	4492.0	0.00	0.49
120-20-sif-3	4473.0	6.61	180.41	120-20-sif-3	4464.0	0.00	7.93	120-20-sif-3	4461.0	0.00	20.78
120-20-sif-4	4393.0	0.00	3.00	120-20-sif-4	4392.0	0.00	0.91	120-20-sif-4	4385.0	0.00	0.07
120-20-sif-5	4450.0	0.00	65.38	120-20-sif-5	4446.0	0.00	0.79	120-20-sif-5	4446.0	0.00	0.07
120-30-sif-1	6658.0	1.75	180.55	120-30-sif-1	6655.0	0.00	0.23	120-30-sif-1	6655.0	0.00	0.73
120-30-sif-2	6726.0	0.00	73.81	120-30-sif-2	6724.0	0.00	48.28	120-30-sif-2	6722.0	0.00	0.30
120-30-sif-3	6634.0	0.00	5.15	120-30-sif-3	6633.0	0.00	0.54	120-30-sif-3	6633.0	0.00	0.50
120-30-sif-4	6640.0	0.00	180.24	120-30-sif-4	6639.0	1.51	180.28	120-30-sif-4	6639.0	1.51	180.31
120-30-sif-5	6566.0	0.00	27.49	120-30-sif-5	6563.0	0.00	6.94	120-30-sif-5	6563.0	0.00	1.61

2. Detailed Results of Standard-Weight Instances on S_A under WTFRPA Setting

We first report the results of DK23 on the instance with $ws = 0.3$. Among these instances, only the instance "60-10-sif-0.3-3" is feasible.

Table 8 Detailed Results of DK23 on 60-10-sif-0.3-3

ws-0.3			
Name	Cost	Gap‰	CPU
60-10-sif-3	2301.0	0.00	0.01

2.1. Results on the instances with i execution mode

Table 9 Detailed Results of DK23 on Standard-Weight Instances in S_A with i execution mode ($ws = 0.4 - 0.6$)

ws-0.4				ws-0.5				ws-0.6			
Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU
60-10-i-1	-	-	-	60-10-i-1	-	-	-	60-10-i-1	-	-	-
60-10-i-2	-	-	-	60-10-i-2	-	-	-	60-10-i-2	-	-	-
60-10-i-3	-	-	-	60-10-i-3	-	-	-	60-10-i-3	2303.0	0.00	0.06
60-10-i-4	-	-	-	60-10-i-4	-	-	-	60-10-i-4	-	-	-
60-10-i-5	-	-	-	60-10-i-5	-	-	-	60-10-i-5	-	-	-
60-20-i-1	-	-	-	60-20-i-1	-	-	-	60-20-i-1	-	-	-
60-20-i-2	-	-	-	60-20-i-2	-	-	-	60-20-i-2	-	-	-
60-20-i-3	-	-	-	60-20-i-3	-	-	-	60-20-i-3	-	-	-
60-20-i-4	-	-	-	60-20-i-4	-	-	-	60-20-i-4	-	-	-
60-20-i-5	-	-	-	60-20-i-5	-	-	-	60-20-i-5	-	-	-
60-30-i-1	-	-	-	60-30-i-1	-	-	-	60-30-i-1	-	-	-
60-30-i-2	-	-	-	60-30-i-2	-	-	-	60-30-i-2	-	-	-
60-30-i-3	-	-	-	60-30-i-3	-	-	-	60-30-i-3	-	-	-
60-30-i-4	-	-	-	60-30-i-4	-	-	-	60-30-i-4	-	-	-
60-30-i-5	-	-	-	60-30-i-5	-	-	-	60-30-i-5	-	-	-
90-10-i-1	-	-	-	90-10-i-1	-	-	-	90-10-i-1	-	-	-
90-10-i-2	-	-	-	90-10-i-2	-	-	-	90-10-i-2	-	-	-
90-10-i-3	-	-	-	90-10-i-3	-	-	-	90-10-i-3	-	-	-
90-10-i-4	-	-	-	90-10-i-4	-	-	-	90-10-i-4	-	-	-
90-10-i-5	-	-	-	90-10-i-5	-	-	-	90-10-i-5	-	-	-
90-20-i-1	-	-	-	90-20-i-1	-	-	-	90-20-i-1	-	-	-
90-20-i-2	-	-	-	90-20-i-2	-	-	-	90-20-i-2	-	-	-
90-20-i-3	-	-	-	90-20-i-3	-	-	-	90-20-i-3	-	-	-
90-20-i-4	-	-	-	90-20-i-4	-	-	-	90-20-i-4	-	-	-
90-20-i-5	-	-	-	90-20-i-5	-	-	-	90-20-i-5	-	-	-
90-30-i-1	-	-	-	90-30-i-1	-	-	-	90-30-i-1	-	-	-
90-30-i-2	-	-	-	90-30-i-2	-	-	-	90-30-i-2	-	-	-
90-30-i-3	-	-	-	90-30-i-3	-	-	-	90-30-i-3	-	-	-
90-30-i-4	-	-	-	90-30-i-4	-	-	-	90-30-i-4	-	-	-
90-30-i-5	-	-	-	90-30-i-5	-	-	-	90-30-i-5	-	-	-
120-10-i-1	-	-	-	120-10-i-1	-	-	-	120-10-i-1	-	-	-
120-10-i-2	-	-	-	120-10-i-2	-	-	-	120-10-i-2	-	-	-
120-10-i-3	-	-	-	120-10-i-3	-	-	-	120-10-i-3	-	-	-
120-10-i-4	-	-	-	120-10-i-4	-	-	-	120-10-i-4	-	-	-
120-10-i-5	-	-	-	120-10-i-5	-	-	-	120-10-i-5	-	-	-
120-20-i-1	-	-	-	120-20-i-1	-	-	-	120-20-i-1	-	-	-
120-20-i-2	-	-	-	120-20-i-2	-	-	-	120-20-i-2	-	-	-
120-20-i-3	-	-	-	120-20-i-3	-	-	-	120-20-i-3	-	-	-
120-20-i-4	-	-	-	120-20-i-4	-	-	-	120-20-i-4	-	-	-
120-20-i-5	-	-	-	120-20-i-5	-	-	-	120-20-i-5	-	-	-
120-30-i-1	-	-	-	120-30-i-1	-	-	-	120-30-i-1	-	-	-
120-30-i-2	-	-	-	120-30-i-2	-	-	-	120-30-i-2	12026.0	7.17	180.17
120-30-i-3	-	-	-	120-30-i-3	-	-	-	120-30-i-3	11254.0	1.70	180.31
120-30-i-4	-	-	-	120-30-i-4	-	-	-	120-30-i-4	-	-	-
120-30-i-5	-	-	-	120-30-i-5	-	-	-	120-30-i-5	-	-	-

Table 10 Detailed Results of DK23 on Standard-Weight Instances in S_A with i execution mode ($ws = 0.7 - 0.9$)

ws-0.7				ws-0.8				ws-0.9			
Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU
60-10-i-1	-	-	-	60-10-i-1	1881.0	0.00	0.01	60-10-i-1	1879.0	0.00	0.01
60-10-i-2	2019.0	0.00	0.09	60-10-i-2	2016.0	0.00	0.07	60-10-i-2	2011.0	0.00	0.01
60-10-i-3	2289.0	0.00	0.01	60-10-i-3	2289.0	0.00	0.01	60-10-i-3	2289.0	0.00	0.01
60-10-i-4	-	-	-	60-10-i-4	-	-	-	60-10-i-4	-	-	-
60-10-i-5	-	-	-	60-10-i-5	-	-	-	60-10-i-5	1925.0	0.00	0.02
60-20-i-1	3774.0	0.00	0.23	60-20-i-1	3770.0	0.00	0.02	60-20-i-1	3770.0	0.00	0.01
60-20-i-2	3954.0	0.00	2.44	60-20-i-2	3948.0	0.00	0.14	60-20-i-2	3946.0	0.00	0.01
60-20-i-3	3200.0	0.00	0.07	60-20-i-3	3198.0	0.00	0.01	60-20-i-3	3196.0	0.00	0.01
60-20-i-4	-	-	-	60-20-i-4	-	-	-	60-20-i-4	3844.0	0.00	0.01
60-20-i-5	-	-	-	60-20-i-5	3658.0	0.00	0.05	60-20-i-5	3658.0	0.00	0.02
60-30-i-1	-	-	-	60-30-i-1	6732.0	0.00	0.08	60-30-i-1	6725.0	0.00	0.24
60-30-i-2	-	-	-	60-30-i-2	-	-	-	60-30-i-2	5786.0	0.00	0.03
60-30-i-3	-	-	-	60-30-i-3	5340.0	0.00	0.03	60-30-i-3	5338.0	0.00	0.02
60-30-i-4	-	-	-	60-30-i-4	5538.0	0.00	0.35	60-30-i-4	5533.0	0.00	0.09
60-30-i-5	-	-	-	60-30-i-5	-	-	-	60-30-i-5	5328.0	0.00	0.02
90-10-i-1	-	-	-	90-10-i-1	2950.0	0.00	0.03	90-10-i-1	2950.0	0.00	0.03
90-10-i-2	-	-	-	90-10-i-2	3024.0	0.00	0.06	90-10-i-2	3024.0	0.00	0.18
90-10-i-3	2948.0	0.00	0.02	90-10-i-3	2948.0	0.00	0.13	90-10-i-3	2944.0	0.00	0.01
90-10-i-4	-	-	-	90-10-i-4	-	-	-	90-10-i-4	3186.0	0.00	0.02
90-10-i-5	-	-	-	90-10-i-5	-	-	-	90-10-i-5	2641.0	0.00	0.01
90-20-i-1	-	-	-	90-20-i-1	6565.0	0.00	0.18	90-20-i-1	6565.0	0.00	0.12
90-20-i-2	-	-	-	90-20-i-2	6877.0	0.00	15.65	90-20-i-2	6873.0	0.00	0.05
90-20-i-3	-	-	-	90-20-i-3	6184.0	0.00	0.25	90-20-i-3	6175.0	0.00	0.03
90-20-i-4	5311.0	4.00	180.02	90-20-i-4	5303.0	0.00	0.07	90-20-i-4	5301.0	0.00	0.02
90-20-i-5	-	-	-	90-20-i-5	6029.0	0.00	0.08	90-20-i-5	6025.0	0.00	0.08
90-30-i-1	-	-	-	90-30-i-1	9625.0	0.00	0.34	90-30-i-1	9625.0	0.00	0.31
90-30-i-2	-	-	-	90-30-i-2	-	-	-	90-30-i-2	8581.0	0.00	0.08
90-30-i-3	-	-	-	90-30-i-3	-	-	-	90-30-i-3	8719.0	0.00	0.06
90-30-i-4	-	-	-	90-30-i-4	9026.0	0.00	0.12	90-30-i-4	9026.0	0.00	0.11
90-30-i-5	-	-	-	90-30-i-5	8040.0	0.00	0.11	90-30-i-5	8040.0	0.00	0.07
120-10-i-1	3892.0	7.55	180.02	120-10-i-1	3890.0	3.86	180.02	120-10-i-1	3886.0	0.00	0.03
120-10-i-2	-	-	-	120-10-i-2	-	-	-	120-10-i-2	3571.0	0.00	0.08
120-10-i-3	-	-	-	120-10-i-3	-	-	-	120-10-i-3	3440.0	0.00	0.18
120-10-i-4	-	-	-	120-10-i-4	4352.0	0.00	0.04	120-10-i-4	4352.0	0.00	0.14
120-10-i-5	-	-	-	120-10-i-5	-	-	-	120-10-i-5	3858.0	0.00	0.06
120-20-i-1	-	-	-	120-20-i-1	-	-	-	120-20-i-1	8143.0	0.00	0.10
120-20-i-2	-	-	-	120-20-i-2	-	-	-	120-20-i-2	9375.0	0.00	0.06
120-20-i-3	-	-	-	120-20-i-3	9742.0	0.00	0.07	120-20-i-3	9742.0	0.00	0.19
120-20-i-4	-	-	-	120-20-i-4	-	-	-	120-20-i-4	7792.0	0.00	0.96
120-20-i-5	7534.0	1.33	180.03	120-20-i-5	7532.0	0.00	1.01	120-20-i-5	7532.0	0.00	0.16
120-30-i-1	13095.0	3.05	180.13	120-30-i-1	13086.0	1.81	180.12	120-30-i-1	13080.0	0.00	3.26
120-30-i-2	12008.0	0.00	2.07	120-30-i-2	12002.0	0.00	1.67	120-30-i-2	12002.0	0.00	0.41
120-30-i-3	11246.0	0.00	0.22	120-30-i-3	11246.0	0.00	2.90	120-30-i-3	11246.0	0.00	0.68
120-30-i-4	-	-	-	120-30-i-4	11498.0	0.00	0.15	120-30-i-4	11490.0	0.00	0.14
120-30-i-5	-	-	-	120-30-i-5	-	-	-	120-30-i-5	13444.0	0.00	0.67

2.2. Results on the instances with sf execution mode

Table 11 Detailed Results of DK23 on Standard-Weight Instances in S_A with sf execution mode ($ws = 0.4 - 0.6$)

ws-0.4				ws-0.5				ws-0.6			
Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU
60-10-sf-1	1884.0	0.00	2.19	60-10-sf-1	1868.0	0.00	0.06	60-10-sf-1	1864.0	0.00	0.25
60-10-sf-2	-	-	-	60-10-sf-2	-	-	-	60-10-sf-2	1998.0	0.00	80.07
60-10-sf-3	2283.0	0.00	0.12	60-10-sf-3	2269.0	0.00	0.10	60-10-sf-3	2265.0	0.00	0.04
60-10-sf-4	-	-	-	60-10-sf-4	2037.0	0.00	0.02	60-10-sf-4	2019.0	0.00	0.23
60-10-sf-5	-	-	-	60-10-sf-5	1919.0	0.00	0.02	60-10-sf-5	1910.0	0.00	0.01
60-20-sf-1	-	-	-	60-20-sf-1	3757.0	0.00	0.03	60-20-sf-1	3747.0	0.00	0.27
60-20-sf-2	-	-	-	60-20-sf-2	-	-	-	60-20-sf-2	3937.0	0.00	0.13
60-20-sf-3	-	-	-	60-20-sf-3	-	-	-	60-20-sf-3	-	-	-
60-20-sf-4	-	-	-	60-20-sf-4	3831.0	0.00	0.13	60-20-sf-4	3814.0	0.00	0.03
60-20-sf-5	-	-	-	60-20-sf-5	3651.0	0.00	0.06	60-20-sf-5	3637.0	0.00	0.10
60-30-sf-1	-	-	-	60-30-sf-1	6684.0	0.00	0.32	60-30-sf-1	6660.0	0.00	0.08
60-30-sf-2	-	-	-	60-30-sf-2	-	-	-	60-30-sf-2	5742.0	2.10	180.71
60-30-sf-3	-	-	-	60-30-sf-3	-	-	-	60-30-sf-3	5307.0	0.00	6.90
60-30-sf-4	-	-	-	60-30-sf-4	5527.0	3.62	180.07	60-30-sf-4	5496.0	0.00	13.14
60-30-sf-5	-	-	-	60-30-sf-5	-	-	-	60-30-sf-5	5293.0	0.00	6.20
90-10-sf-1	-	-	-	90-10-sf-1	-	-	-	90-10-sf-1	2921.0	0.00	1.28
90-10-sf-2	-	-	-	90-10-sf-2	-	-	-	90-10-sf-2	3002.0	0.00	0.02
90-10-sf-3	-	-	-	90-10-sf-3	2921.0	0.00	0.07	90-10-sf-3	2916.0	0.00	0.02
90-10-sf-4	-	-	-	90-10-sf-4	-	-	-	90-10-sf-4	-	-	-
90-10-sf-5	2639.0	11.37	180.06	90-10-sf-5	2626.0	0.00	0.08	90-10-sf-5	2617.0	0.00	0.24
90-20-sf-1	-	-	-	90-20-sf-1	6525.0	1.53	180.06	90-20-sf-1	6501.0	0.00	180.08
90-20-sf-2	-	-	-	90-20-sf-2	6808.0	0.00	13.86	90-20-sf-2	6786.0	4.42	180.11
90-20-sf-3	-	-	-	90-20-sf-3	6136.0	0.00	0.54	90-20-sf-3	6115.0	0.00	176.68
90-20-sf-4	-	-	-	90-20-sf-4	-	-	-	90-20-sf-4	5274.0	3.79	180.37
90-20-sf-5	-	-	-	90-20-sf-5	-	-	-	90-20-sf-5	5977.0	5.02	180.06
90-30-sf-1	-	-	-	90-30-sf-1	-	-	-	90-30-sf-1	-	-	-
90-30-sf-2	-	-	-	90-30-sf-2	-	-	-	90-30-sf-2	-	-	-
90-30-sf-3	-	-	-	90-30-sf-3	8671.0	3.85	181.39	90-30-sf-3	8647.0	1.33	180.37
90-30-sf-4	-	-	-	90-30-sf-4	9003.0	0.00	1.92	90-30-sf-4	8954.0	1.12	180.85
90-30-sf-5	-	-	-	90-30-sf-5	7993.0	0.00	1.31	90-30-sf-5	7975.0	6.27	180.26
120-10-sf-1	-	-	-	120-10-sf-1	-	-	-	120-10-sf-1	3855.0	0.00	1.18
120-10-sf-2	-	-	-	120-10-sf-2	-	-	-	120-10-sf-2	3555.0	0.00	0.04
120-10-sf-3	-	-	-	120-10-sf-3	-	-	-	120-10-sf-3	3413.0	0.00	1.28
120-10-sf-4	-	-	-	120-10-sf-4	4317.0	0.00	3.10	120-10-sf-4	4315.0	0.00	0.60
120-10-sf-5	-	-	-	120-10-sf-5	-	-	-	120-10-sf-5	3832.0	0.00	0.15
120-20-sf-1	8096.0	12.25	180.11	120-20-sf-1	8080.0	23.51	180.24	120-20-sf-1	8055.0	0.00	40.39
120-20-sf-2	-	-	-	120-20-sf-2	9326.0	16.16	180.24	120-20-sf-2	9282.0	0.00	2.92
120-20-sf-3	-	-	-	120-20-sf-3	-	-	-	120-20-sf-3	9633.0	1.82	180.43
120-20-sf-4	-	-	-	120-20-sf-4	-	-	-	120-20-sf-4	-	-	-
120-20-sf-5	-	-	-	120-20-sf-5	-	-	-	120-20-sf-5	-	-	-
120-30-sf-1	-	-	-	120-30-sf-1	13016.0	7.12	180.21	120-30-sf-1	12965.0	0.00	2.34
120-30-sf-2	-	-	-	120-30-sf-2	11939.0	6.04	181.29	120-30-sf-2	11901.0	2.32	181.31
120-30-sf-3	-	-	-	120-30-sf-3	11214.0	35.45	180.82	120-30-sf-3	11156.0	0.00	3.12
120-30-sf-4	-	-	-	120-30-sf-4	11439.0	10.34	184.44	120-30-sf-4	11397.0	5.02	181.39
120-30-sf-5	-	-	-	120-30-sf-5	-	-	-	120-30-sf-5	-	-	-

Table 12 Detailed Results of DK23 on Standard-Weight Instances in S_A with sf execution mode ($ws = 0.7 - 0.9$)

ws-0.7				ws-0.8				ws-0.9			
Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU
60-10-sf-1	1859.0	0.00	0.01	60-10-sf-1	1859.0	0.00	0.01	60-10-sf-1	1859.0	0.00	0.01
60-10-sf-2	1992.0	0.00	1.56	60-10-sf-2	1992.0	0.00	1.55	60-10-sf-2	1989.0	0.00	0.01
60-10-sf-3	2261.0	0.00	0.02	60-10-sf-3	2251.0	0.00	0.01	60-10-sf-3	2251.0	0.00	0.01
60-10-sf-4	2011.0	0.00	0.04	60-10-sf-4	2009.0	0.00	0.02	60-10-sf-4	2007.0	0.00	0.01
60-10-sf-5	1906.0	0.00	0.01	60-10-sf-5	1904.0	0.00	0.04	60-10-sf-5	1900.0	0.00	0.01
60-20-sf-1	3742.0	0.00	0.42	60-20-sf-1	3738.0	0.00	0.02	60-20-sf-1	3738.0	0.00	0.02
60-20-sf-2	3920.0	0.00	0.05	60-20-sf-2	3904.0	0.00	0.10	60-20-sf-2	3902.0	0.00	0.03
60-20-sf-3	3182.0	0.00	0.04	60-20-sf-3	3180.0	0.00	0.05	60-20-sf-3	3175.0	0.00	0.02
60-20-sf-4	3809.0	0.00	0.06	60-20-sf-4	3806.0	0.00	0.12	60-20-sf-4	3804.0	0.00	0.08
60-20-sf-5	3630.0	0.00	0.08	60-20-sf-5	3628.0	0.00	0.08	60-20-sf-5	3626.0	0.00	0.02
60-30-sf-1	6651.0	0.00	0.16	60-30-sf-1	6644.0	0.00	0.17	60-30-sf-1	6642.0	0.00	0.10
60-30-sf-2	5711.0	0.00	7.76	60-30-sf-2	5705.0	0.00	1.00	60-30-sf-2	5704.0	0.00	0.04
60-30-sf-3	5289.0	0.00	0.49	60-30-sf-3	5280.0	0.00	4.25	60-30-sf-3	5277.0	0.00	3.42
60-30-sf-4	5482.0	0.00	0.24	60-30-sf-4	5476.0	0.00	0.69	60-30-sf-4	5472.0	0.00	0.17
60-30-sf-5	5280.0	0.00	1.18	60-30-sf-5	5277.0	2.08	180.09	60-30-sf-5	5273.0	0.00	0.33
90-10-sf-1	2920.0	8.13	180.03	90-10-sf-1	2912.0	15.45	180.02	90-10-sf-1	2912.0	15.45	180.03
90-10-sf-2	2995.0	0.00	0.47	90-10-sf-2	2989.0	0.00	0.09	90-10-sf-2	2988.0	0.00	0.03
90-10-sf-3	2915.0	0.00	0.04	90-10-sf-3	2915.0	0.00	0.08	90-10-sf-3	2915.0	0.00	0.03
90-10-sf-4	-	-	-	90-10-sf-4	3153.0	0.00	0.06	90-10-sf-4	3153.0	0.00	0.07
90-10-sf-5	2613.0	0.00	0.20	90-10-sf-5	2612.0	0.00	0.04	90-10-sf-5	2612.0	0.00	0.10
90-20-sf-1	6491.0	0.00	0.19	90-20-sf-1	6487.0	0.00	0.08	90-20-sf-1	6487.0	0.00	0.27
90-20-sf-2	6784.0	6.26	180.13	90-20-sf-2	6781.0	4.42	180.12	90-20-sf-2	6775.0	0.00	0.99
90-20-sf-3	6109.0	2.35	180.09	90-20-sf-3	6107.0	0.00	0.47	90-20-sf-3	6107.0	0.00	0.43
90-20-sf-4	5262.0	1.90	180.15	90-20-sf-4	5255.0	0.00	0.70	90-20-sf-4	5252.0	0.00	0.83
90-20-sf-5	5967.0	3.35	180.11	90-20-sf-5	5955.0	0.00	0.13	90-20-sf-5	5952.0	0.00	0.07
90-30-sf-1	9535.0	1.05	180.19	90-30-sf-1	9502.0	0.00	180.12	90-30-sf-1	9498.0	0.00	0.36
90-30-sf-2	8506.0	4.41	180.17	90-30-sf-2	8495.0	5.89	180.13	90-30-sf-2	8485.0	1.77	180.13
90-30-sf-3	8638.0	1.54	180.27	90-30-sf-3	8634.0	1.16	180.37	90-30-sf-3	8628.0	0.00	0.70
90-30-sf-4	8935.0	2.24	180.21	90-30-sf-4	8931.0	0.00	0.78	90-30-sf-4	8929.0	0.00	0.45
90-30-sf-5	7963.0	0.00	0.62	90-30-sf-5	7961.0	0.00	14.34	90-30-sf-5	7959.0	0.00	0.09
120-10-sf-1	3852.0	0.00	0.72	120-10-sf-1	3850.0	0.00	0.06	120-10-sf-1	3850.0	0.00	0.04
120-10-sf-2	3554.0	0.00	0.33	120-10-sf-2	3551.0	0.00	0.08	120-10-sf-2	3551.0	0.00	0.10
120-10-sf-3	3411.0	6.60	180.16	120-10-sf-3	3409.0	5.13	180.07	120-10-sf-3	3409.0	5.87	180.07
120-10-sf-4	4311.0	3.48	180.14	120-10-sf-4	4307.0	0.00	0.33	120-10-sf-4	4289.0	0.00	0.04
120-10-sf-5	3828.0	0.00	15.00	120-10-sf-5	3821.0	0.00	0.30	120-10-sf-5	3821.0	0.00	0.18
120-20-sf-1	8051.0	0.00	0.31	120-20-sf-1	8043.0	0.00	0.15	120-20-sf-1	8043.0	0.00	0.27
120-20-sf-2	9264.0	0.00	0.48	120-20-sf-2	9255.0	0.00	7.34	120-20-sf-2	9246.0	0.00	0.21
120-20-sf-3	9610.0	3.90	181.15	120-20-sf-3	9596.0	0.00	14.02	120-20-sf-3	9590.0	1.56	180.17
120-20-sf-4	-	-	-	120-20-sf-4	7697.0	0.00	0.07	120-20-sf-4	7695.0	0.00	0.85
120-20-sf-5	-	-	-	120-20-sf-5	7455.0	0.00	1.64	120-20-sf-5	7455.0	0.00	0.25
120-30-sf-1	12956.0	0.00	1.40	120-30-sf-1	12954.0	0.00	0.82	120-30-sf-1	12954.0	0.00	0.12
120-30-sf-2	11889.0	1.68	180.78	120-30-sf-2	11884.0	0.84	180.48	120-30-sf-2	11879.0	0.00	7.11
120-30-sf-3	11152.0	0.00	1.59	120-30-sf-3	11152.0	0.00	3.46	120-30-sf-3	11152.0	0.00	1.14
120-30-sf-4	11368.0	1.76	180.21	120-30-sf-4	11366.0	2.64	180.37	120-30-sf-4	11357.0	0.00	0.61
120-30-sf-5	13293.0	0.00	9.50	120-30-sf-5	13287.0	0.00	0.80	120-30-sf-5	13266.0	0.00	3.56

2.3. Results on the instances with sif execution mode

Table 13 Detailed Results of DK23 on Standard-Weight Instances in S_A with sif execution mode ($ws = 0.4 - 0.6$)

ws-0.4				ws-0.5				ws-0.6			
Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU
60-10-sif-1	1880.0	0.00	0.01	60-10-sif-1	1872.0	0.00	0.03	60-10-sif-1	1866.0	0.00	0.06
60-10-sif-2	-	-	-	60-10-sif-2	2021.0	0.00	0.03	60-10-sif-2	2005.0	0.00	0.01
60-10-sif-3	2289.0	0.00	0.07	60-10-sif-3	2271.0	0.00	0.02	60-10-sif-3	2268.0	0.00	0.75
60-10-sif-4	-	-	-	60-10-sif-4	2037.0	0.00	0.01	60-10-sif-4	2019.0	0.00	0.03
60-10-sif-5	1917.0	0.00	0.01	60-10-sif-5	1912.0	0.00	0.05	60-10-sif-5	1908.0	0.00	0.03
60-20-sif-1	-	-	-	60-20-sif-1	3757.0	0.00	0.08	60-20-sif-1	3748.0	0.00	4.75
60-20-sif-2	-	-	-	60-20-sif-2	3939.0	2.54	180.07	60-20-sif-2	3926.0	0.00	0.26
60-20-sif-3	-	-	-	60-20-sif-3	-	-	-	60-20-sif-3	3196.0	0.00	29.97
60-20-sif-4	-	-	-	60-20-sif-4	3826.0	0.00	0.08	60-20-sif-4	3814.0	0.00	0.03
60-20-sif-5	-	-	-	60-20-sif-5	3652.0	0.00	57.11	60-20-sif-5	3639.0	0.00	0.17
60-30-sif-1	-	-	-	60-30-sif-1	6674.0	0.00	2.05	60-30-sif-1	6662.0	0.00	0.05
60-30-sif-2	-	-	-	60-30-sif-2	-	-	-	60-30-sif-2	5745.0	4.22	180.70
60-30-sif-3	-	-	-	60-30-sif-3	-	-	-	60-30-sif-3	5298.0	0.00	1.89
60-30-sif-4	-	-	-	60-30-sif-4	-	-	-	60-30-sif-4	5492.0	0.00	0.67
60-30-sif-5	-	-	-	60-30-sif-5	-	-	-	60-30-sif-5	5297.0	0.00	1.34
90-10-sif-1	-	-	-	90-10-sif-1	2934.0	0.00	173.85	90-10-sif-1	2925.0	6.38	180.09
90-10-sif-2	-	-	-	90-10-sif-2	3017.0	0.00	0.07	90-10-sif-2	2998.0	0.00	0.05
90-10-sif-3	-	-	-	90-10-sif-3	2930.0	0.00	0.02	90-10-sif-3	2917.0	0.00	0.02
90-10-sif-4	-	-	-	90-10-sif-4	-	-	-	90-10-sif-4	-	-	-
90-10-sif-5	-	-	-	90-10-sif-5	2626.0	0.00	0.45	90-10-sif-5	2617.0	0.00	0.45
90-20-sif-1	-	-	-	90-20-sif-1	6538.0	0.00	3.66	90-20-sif-1	6501.0	0.00	180.14
90-20-sif-2	-	-	-	90-20-sif-2	6840.0	5.85	180.72	90-20-sif-2	6800.0	1.47	180.35
90-20-sif-3	-	-	-	90-20-sif-3	6140.0	0.00	2.09	90-20-sif-3	6117.0	0.00	132.51
90-20-sif-4	-	-	-	90-20-sif-4	-	-	-	90-20-sif-4	5273.0	0.00	180.21
90-20-sif-5	-	-	-	90-20-sif-5	-	-	-	90-20-sif-5	5976.0	0.00	0.78
90-30-sif-1	-	-	-	90-30-sif-1	9601.0	7.38	181.63	90-30-sif-1	9552.0	3.49	181.24
90-30-sif-2	-	-	-	90-30-sif-2	-	-	-	90-30-sif-2	8524.0	0.00	1.95
90-30-sif-3	-	-	-	90-30-sif-3	8673.0	0.00	60.92	90-30-sif-3	8648.0	0.00	16.94
90-30-sif-4	-	-	-	90-30-sif-4	8986.0	0.00	10.16	90-30-sif-4	8947.0	0.00	0.66
90-30-sif-5	8049.0	0.00	0.17	90-30-sif-5	7992.0	0.00	6.67	90-30-sif-5	7971.0	1.88	180.94
120-10-sif-1	-	-	-	120-10-sif-1	-	-	-	120-10-sif-1	3868.0	10.34	180.13
120-10-sif-2	3567.0	0.00	48.67	120-10-sif-2	3557.0	0.00	0.18	120-10-sif-2	3555.0	0.00	0.21
120-10-sif-3	-	-	-	120-10-sif-3	-	-	-	120-10-sif-3	3416.0	2.93	180.26
120-10-sif-4	-	-	-	120-10-sif-4	4319.0	0.00	84.06	120-10-sif-4	4307.0	0.00	0.28
120-10-sif-5	-	-	-	120-10-sif-5	-	-	-	120-10-sif-5	3834.0	0.00	44.58
120-20-sif-1	-	-	-	120-20-sif-1	8066.0	0.00	1.20	120-20-sif-1	8055.0	0.00	180.18
120-20-sif-2	-	-	-	120-20-sif-2	9318.0	0.00	6.01	120-20-sif-2	9276.0	0.00	0.44
120-20-sif-3	-	-	-	120-20-sif-3	-	-	-	120-20-sif-3	9643.0	0.00	60.27
120-20-sif-4	-	-	-	120-20-sif-4	7744.0	6.13	180.36	120-20-sif-4	7727.0	4.93	180.36
120-20-sif-5	-	-	-	120-20-sif-5	-	-	-	120-20-sif-5	7472.0	0.00	5.35
120-30-sif-1	-	-	-	120-30-sif-1	13058.0	5.36	181.32	120-30-sif-1	12975.0	2.06	180.96
120-30-sif-2	-	-	-	120-30-sif-2	11925.0	1.82	181.15	120-30-sif-2	11898.0	0.00	1.56
120-30-sif-3	-	-	-	120-30-sif-3	11195.0	0.00	181.73	120-30-sif-3	11157.0	0.00	4.05
120-30-sif-4	-	-	-	120-30-sif-4	11455.0	5.76	183.91	120-30-sif-4	11390.0	0.00	180.66
120-30-sif-5	-	-	-	120-30-sif-5	-	-	-	120-30-sif-5	13316.0	1.44	182.03

Table 14 Detailed Results of DK23 on Standard-Weight Instances in S_A with sif execution mode ($ws = 0.7 - 0.9$)

ws-0.7				ws-0.8				ws-0.9			
Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU
60-10-sif-1	1862.0	0.00	0.23	60-10-sif-1	1859.0	0.00	0.05	60-10-sif-1	1859.0	0.00	0.05
60-10-sif-2	2003.0	0.00	0.18	60-10-sif-2	1992.0	0.00	3.70	60-10-sif-2	1989.0	0.00	0.01
60-10-sif-3	2263.0	0.00	0.04	60-10-sif-3	2260.0	0.00	0.01	60-10-sif-3	2251.0	0.00	0.02
60-10-sif-4	2011.0	0.00	0.07	60-10-sif-4	2009.0	0.00	0.02	60-10-sif-4	2007.0	0.00	0.02
60-10-sif-5	1904.0	0.00	0.04	60-10-sif-5	1902.0	0.00	0.01	60-10-sif-5	1900.0	0.00	0.02
60-20-sif-1	3742.0	0.00	0.06	60-20-sif-1	3738.0	0.00	0.02	60-20-sif-1	3738.0	0.00	0.11
60-20-sif-2	3919.0	0.00	0.25	60-20-sif-2	3912.0	0.00	0.18	60-20-sif-2	3902.0	0.00	0.08
60-20-sif-3	3178.0	0.00	0.03	60-20-sif-3	3178.0	0.00	0.03	60-20-sif-3	3175.0	0.00	0.01
60-20-sif-4	3811.0	0.00	0.08	60-20-sif-4	3806.0	0.00	0.10	60-20-sif-4	3804.0	0.00	0.23
60-20-sif-5	3630.0	0.00	0.38	60-20-sif-5	3628.0	0.00	0.15	60-20-sif-5	3626.0	0.00	0.02
60-30-sif-1	6651.0	0.00	1.02	60-30-sif-1	6639.0	0.00	0.13	60-30-sif-1	6636.0	0.00	0.10
60-30-sif-2	5714.0	0.00	5.20	60-30-sif-2	5705.0	0.00	0.86	60-30-sif-2	5704.0	0.00	0.05
60-30-sif-3	5289.0	0.00	37.14	60-30-sif-3	5283.0	0.00	0.10	60-30-sif-3	5277.0	0.00	0.13
60-30-sif-4	5483.0	4.86	180.07	60-30-sif-4	5476.0	1.83	180.07	60-30-sif-4	5472.0	0.00	0.07
60-30-sif-5	5283.0	1.89	180.11	60-30-sif-5	5277.0	0.00	1.69	60-30-sif-5	5273.0	0.00	0.07
90-10-sif-1	2917.0	8.57	180.04	90-10-sif-1	2915.0	10.29	180.04	90-10-sif-1	2907.0	0.00	0.06
90-10-sif-2	2992.0	0.00	0.09	90-10-sif-2	2989.0	0.00	0.23	90-10-sif-2	2988.0	0.00	0.03
90-10-sif-3	2915.0	0.00	0.03	90-10-sif-3	2915.0	0.00	0.03	90-10-sif-3	2915.0	0.00	0.02
90-10-sif-4	3160.0	0.00	0.04	90-10-sif-4	3159.0	0.00	0.11	90-10-sif-4	3153.0	0.00	0.09
90-10-sif-5	2616.0	0.00	0.05	90-10-sif-5	2612.0	0.00	0.04	90-10-sif-5	2612.0	0.00	0.08
90-20-sif-1	6491.0	0.00	0.57	90-20-sif-1	6487.0	0.00	0.06	90-20-sif-1	6487.0	0.00	0.07
90-20-sif-2	6784.0	6.26	180.21	90-20-sif-2	6781.0	4.42	180.15	90-20-sif-2	6775.0	0.00	0.08
90-20-sif-3	6110.0	0.00	0.97	90-20-sif-3	6101.0	0.00	0.14	90-20-sif-3	6101.0	0.00	0.22
90-20-sif-4	5261.0	0.00	32.53	90-20-sif-4	5255.0	0.00	0.69	90-20-sif-4	5252.0	0.00	0.51
90-20-sif-5	5962.0	0.00	1.16	90-20-sif-5	5955.0	0.00	0.85	90-20-sif-5	5952.0	0.00	0.31
90-30-sif-1	9529.0	0.00	29.12	90-30-sif-1	9523.0	0.00	0.70	90-30-sif-1	9498.0	0.00	0.28
90-30-sif-2	8502.0	5.29	180.57	90-30-sif-2	8494.0	4.71	180.18	90-30-sif-2	8491.0	4.42	180.16
90-30-sif-3	8640.0	1.74	180.37	90-30-sif-3	8634.0	1.16	180.36	90-30-sif-3	8628.0	0.00	0.22
90-30-sif-4	8935.0	0.00	0.20	90-30-sif-4	8931.0	0.00	1.23	90-30-sif-4	8929.0	0.00	0.30
90-30-sif-5	7963.0	0.00	2.21	90-30-sif-5	7961.0	0.00	3.01	90-30-sif-5	7959.0	0.00	0.13
120-10-sif-1	3866.0	15.52	180.19	120-10-sif-1	3850.0	0.00	0.04	120-10-sif-1	3850.0	0.00	0.03
120-10-sif-2	3552.0	0.00	0.93	120-10-sif-2	3549.0	0.00	0.17	120-10-sif-2	3549.0	0.00	0.07
120-10-sif-3	3408.0	0.00	0.04	120-10-sif-3	3407.0	0.00	0.35	120-10-sif-3	3407.0	0.00	0.04
120-10-sif-4	4307.0	0.00	0.35	120-10-sif-4	4305.0	0.00	0.12	120-10-sif-4	4289.0	0.00	0.09
120-10-sif-5	3823.0	0.00	0.26	120-10-sif-5	3819.0	0.00	0.15	120-10-sif-5	3819.0	0.00	0.22
120-20-sif-1	8051.0	0.00	0.58	120-20-sif-1	8049.0	0.00	0.41	120-20-sif-1	8043.0	0.00	0.22
120-20-sif-2	9264.0	0.00	1.01	120-20-sif-2	9255.0	0.00	0.34	120-20-sif-2	9246.0	0.00	0.33
120-20-sif-3	9618.0	4.95	180.26	120-20-sif-3	9598.0	0.00	9.28	120-20-sif-3	9590.0	1.56	180.20
120-20-sif-4	7718.0	1.94	180.35	120-20-sif-4	7716.0	1.94	180.19	120-20-sif-4	7695.0	0.00	1.11
120-20-sif-5	7460.0	0.00	18.31	120-20-sif-5	7455.0	0.00	2.21	120-20-sif-5	7455.0	0.00	0.35
120-30-sif-1	12960.0	1.54	180.80	120-30-sif-1	12954.0	0.00	9.23	120-30-sif-1	12954.0	0.00	2.13
120-30-sif-2	11887.0	0.00	4.00	120-30-sif-2	11884.0	0.84	180.44	120-30-sif-2	11879.0	0.00	5.66
120-30-sif-3	11154.0	0.00	6.19	120-30-sif-3	11152.0	0.00	0.29	120-30-sif-3	11152.0	0.00	3.10
120-30-sif-4	11368.0	1.76	180.41	120-30-sif-4	11366.0	2.64	180.37	120-30-sif-4	11366.0	2.64	180.28
120-30-sif-5	13287.0	0.00	24.21	120-30-sif-5	13281.0	0.00	6.26	120-30-sif-5	13281.0	0.00	0.52